Campus Commuting Decarbonisation: Electric Shuttle Route Optimization Study

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CAMPUS COMMUTING DECARBONISATION: ELECTRIC SHUTTLE ROUTE OPTIMIZATION STUDY

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Project Code: INIT_ELECTRIC_SHUTTLE (Zero-Emission Electric Campus Transit)

Executing Department: Department of Artificial Intelligence & Data Science (DEP_AIDS) and Campus Administration (DEP_ADMN).

Location: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai.

1. CONTEXT AND LAST-MILE TRANSIT BOTTLENECK

Vivekananda Education Society's Institute of Technology (VESIT Green Campus) in Chembur, Mumbai, accommodates a commuter community of 3,200 individuals. Mobility surveys (DOC_TRN_007) confirmed that 48.0% of students and staff utilize Mumbai's suburban rail network, arriving at Chembur and Tilak Nagar stations (Harbour Line) or Kurla station (Central Line). However, the remaining 2.4 km distance between these stations and the campus gate historically constituted a severe last-mile transit bottleneck. Over 1,200 commuters daily relied on fossil-fueled three-wheeler auto-rickshaws, generating congestion and substantial localized air pollutants.

To solve this challenge and advance institutional target TRG_GREEN_COMMUTE (70.0% low-carbon commuting by 2027), the college chartered initiative INIT_ELECTRIC_SHUTTLE, procuring two dedicated electric passenger vans in January 2024.

2. FLEET SPECIFICATIONS AND SMART SOLAR CHARGING REGIME

The transit fleet comprises two commercial Mahindra Supro EV passenger vans:

- Seating Capacity: 14 passengers + 1 driver.

- Powertrain: 48V AC induction electric traction motor delivering 19.5 kW peak power and 58 Nm torque.

- Battery Pack: 20.0 kWh lithium-ion battery system delivering a verified real-world operational range of 115 km per full charge.

- On-Campus Solar Charging Protocol: Vans recharge at two dedicated 7.4 kW Level-2 AC smart EV charging stations installed in the central parking lot. Charging schedules are coordinated through smart controllers to draw power between 11:00 and 15:00 on weekdays, directly capturing peak rooftop solar photovoltaic generation from Academic Block A (BLD_ACA, INIT_SOLAR_PH1). This guarantees that the shuttle fleet runs on 100% clean, zero-emissions electricity.

3. TEN-MONTH OPERATIONAL PERFORMANCE EVALUATION (JAN - OCT 2024)

Fleet telematics and ticketing logs over the initial 10-month service window established:

- Total Distance Traveled: 24,800 fleet kilometres across Chembur and Kurla feeder routes.

- Commuter Trips Delivered: 38,400 passenger-trips.

- Fleet Energy Consumption: 4,464 kWh of electricity consumed (average fleet efficiency: 0.18 kWh/km).

- Operational Route Reliability: 98.4% on-time departure rate during peak morning (07:45 to 10:30) and evening (16:00 to 18:30) academic rush hours.

4. CARBON EMISSIONS ABATEMENT MODELING

Carbon abatement was modeled by calculating displaced passenger-kilometres against the baseline fossil fuel fleet:

- Baseline Displaced Modes: Displaced trips predominantly substituted shared three-wheeler petrol/CNG auto-rickshaws (emission factor: 0.082 kg CO2e/passenger-km) and private two-wheelers (EF_TRANS_BIKE = 0.042 kg CO2e/passenger-km).

- Gross Avoided Emissions: Displacing 92,160 passenger-km of three-wheeler transit prevented 7,557 kg CO2e; displacing 61,440 passenger-km of two-wheeler transit avoided 2,580 kg CO2e.

- Modal Substitution & Traffic Decongestion: Fleet routing prevented private vehicle idling at the campus gate, yielding an aggregate verified net carbon savings of 36,000 kg CO2e in 2024.

5. FLEET EXPANSION AND CAPACITY RECOMMENDATIONS

Passenger demand consistently exceeds current capacity during morning peak hours (average load factor: 104%, with student queues at Kurla station). The study recommends:

- Procuring two 24-seater electric mini-buses by Q2 2025 under Phase 2 transit planning.

- Tripling daily passenger throughput to 115,000 passenger-trips annually.

- Expanding avoided emissions to 95,000 kg CO2e/year, guaranteeing complete fulfillment of TRG_GREEN_COMMUTE and advancing United Nations SDG 11 and SDG 13.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

*This statutory record is indexed in the EcoClock AI Knowledge Graph repository under Canonical ID **DOC_TRN_015**. All empirical utility telemetry, carbon accounting estimates, and SDG alignments articulated in this document have been reviewed and verified by the VESIT Campus Sustainability Taskforce, Department of Computer Engineering (DEP_CMPN), Department of Information Technology (DEP_INFT), Department of Electronics & Telecommunication (DEP_EXTC), Department of Artificial Intelligence & Data Science (DEP_AIDS), and Campus Facilities (DEP_ADMN). Published by: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai 400074.*